Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE B
Course Code & Title: CS8082 Machine Learning Techniques
Name of the Faculty member: Mr.R.Deiva Nayagam

Innovative Practice Description

- **Unit / Topic:** II / Genetic Models of Evaluation and Learning
- **Course Outcome:** CO2: The Students will able to apply the concept of neural network back propagation algorithm and genetic algorithm for problem solving
- **Topic Learning Outcome:** TLO6: Apply genetic algorithms to a variety of learning tasks and other optimization problems
- **Activity Chosen:** Think Pair Share
- **Justification:**

  Think Pair Share is a co-operative discussion strategy that has three phases. During phase 1(Think) the instructor poses a question and allows the students time to ponder their response. During phase 2 (Pair) the student discuss their thoughts with their designated partner or tablemate. . During phase 3 (Share) the teacher will call upon the groups to have them tell the class what they discussed and think about that topic. This activity increases the collaborative learning skills of the students.

- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

  Initially, the faculty raises the question to analyze the concept of genetic models of evaluation and learning. This activity on genetic models of evaluation and learning will helps the students to think individually about the concept of Lamarckian and Baldwin models or answer to a question. Then he/she pair with other students and discuss about their thoughts with the designated group. Finally he/she share ideas with classmates and builds oral communication skills on the concept of genetic models. It helped them to focus attention and engage students in comprehending the reading material. Also the concept is very essential for everyone to know about the concept of genetic models evaluation and learning.

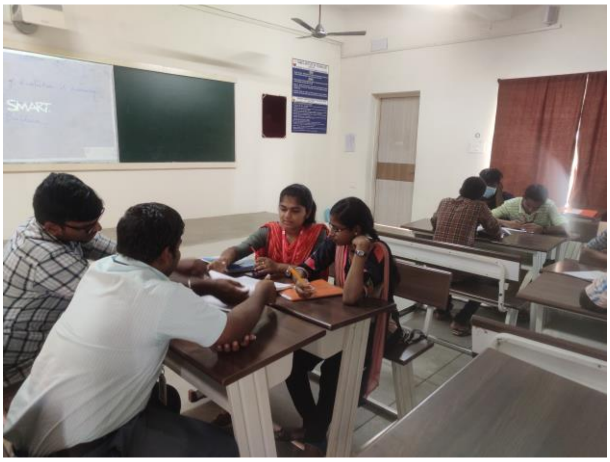
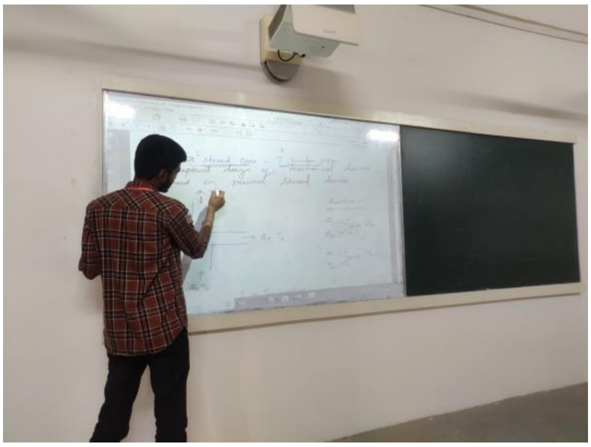
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PSO12	PSO1	PSO3
CO2	3	3	2	2	2	2	2	2	3	2

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PO10**	**PSO1**
	2	2	3
Justification for correlation	To accomplish the think pair share activity, the students implement the norms of practice (rules, roles, agenda) to work individually and form a pair to greater learning.	It helps the students to improve their communication skills and able to explain the problem with the students in a class and make effective presentation.	The concept of artificial genetic models helps to innovate new ideas to solve machine learning problems like face recognition, text classification etc., in various applications.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    After Completion of this activity, students gave feedback that they got a clear idea about the genetic models of evaluation and learning. They felt that it increases the cooperative learning.

  - ***Benefit of the practice:***

    The success of the activity was evaluated by asking this question in IAT II – Around 90% of students answered correctly. Since I conducted this activity, the collaborative learning skills of the students got increased. They actively participated in this activity.

- ***Challenges faced in implementation:***

  Initially, the students feel a little uncomfortable sharing their thoughts on the proposed topic. They are not coming up to be involved in this activity voluntarily. It took a longer time than other strategies.

References:

- https://www.kent.edu/ctl/think-pair-share
- https://www.uopeople.edu/blog/think-pair-share/
- https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html

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Name of the Faculty member: Mr.R.Deiva Nayagam

Innovative Practice Description

- **Unit / Topic:** IV **/** Case Based Reasoning- Instance Based Learning
- **Course Outcome:** CO4: The Students will able to suggest and apply appropriate machine learning algorithms for instance based learning problems
- **Topic Learning Outcome:** TLO13: Apply the concept of Case Based Reasoning learning paradigm for instance based learning
- **Activity Chosen:** JigSaw
- **Justification:**

  The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal; each person depends on all the others.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
- The topic on case based reasoning was divided into sub categories. The Students are divided into groups of four students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PSO12	PSO1
CO4	2	2	2	2	2	2	1	2	3

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PO10**	**PSO1**
	2	1	3
Justification for correlation	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	It helps the students to improve their communication skills and able to explain the problem with the students in a class and make effective presentation.	The concept of instance based learning is efficient way to solve the existing or novel problems in the area of information and communication technology.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students feel that it was interesting. They learn how to communicate with team members and work together.

  - ***Benefit of the practice:***

    Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ***Challenges faced in implementation:***

  The main challenge faced is group conflict. Few Students are not ready to work together. Initially, the students not came up to involve in this activity voluntarily. It took a longer time than other strategies.

References:

- https://www.researchgate.net/publication/318360791_Jigsaw_Method_An_Innovative_Way_of_Cooperative_Learning_in_Physiology
- https://www.readingrockets.org/strategies/jigsaw

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Department of Electronics and Communication Engineering

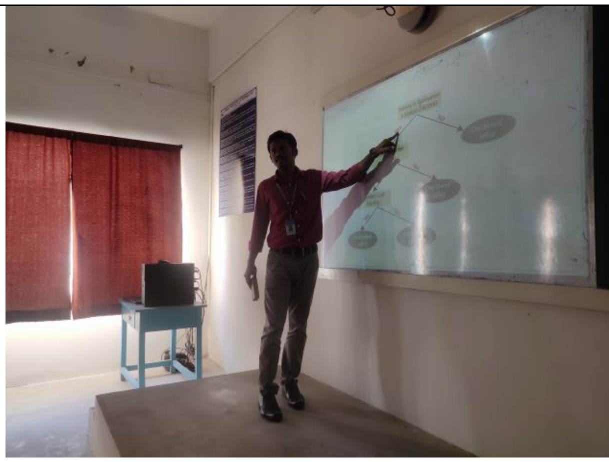
Academic Year 2021 – 2022 (Odd Semester)

Active Learning Description for Unit I

Course Code & Title: CS8082 Machine Learning Techniques
Degree, Semester & Branch: VII Semester B.E. ECE B
Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic	Name of the Innovative Practice
Machine Learning Problems-Perspectives & Issues	Case Studies
C-wise Classification Decision Tree Learning	Problem Based Learning

Various Case Studies discussed about Machine Learning Problems



Problem Based Learning on Decision Tree Algorithm

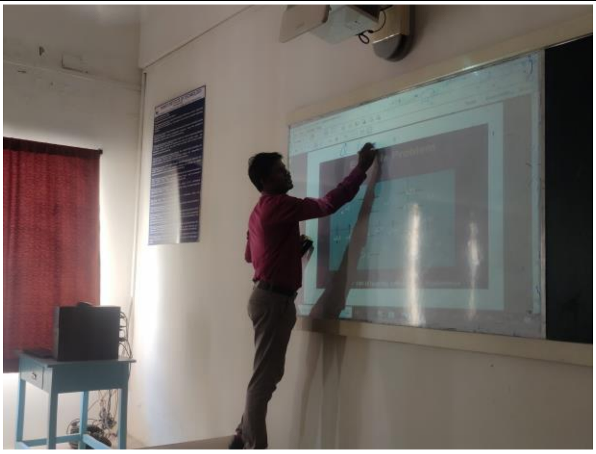
Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Active Learning Description for Unit II

Course Code & Title: CS8082 Machine Learning Techniques
Degree, Semester & Branch: VII Semester B.E. ECE B
Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic	Name of the Innovative Practice
Biological Neural Network, Neural Network Representations	Inquiry based learning
Problem Solving with Neural Network Algorithms	Problem based learning
Back propagation Algorithm	NPTEL Video

Inquiry based learning on Biological Neural Network and its Representations	
	
Problem Based Learning on Neural network Algorithms	
	

NPTEL Video on Back propagation Algorithm	
	

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Department of Electronics and Communication Engineering

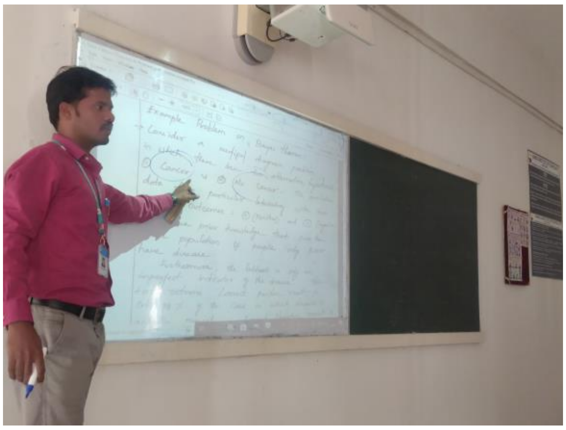
Academic Year 2021 – 2022 (Odd Semester)

Active Learning Description for Unit III

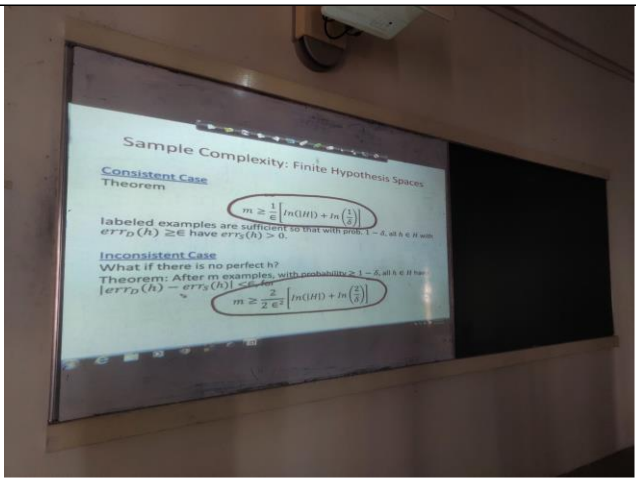
Course Code & Title: CS8082 Machine Learning Techniques
Degree, Semester & Branch: VII Semester B.E. ECE B
Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic	Name of the Innovative Practice
Bayes Theorem and Concept Learning	Problem based learning
Bayesian Belief Networks and EM Algorithm	Problem based learning
Sample Complexity for Finite and Infinite Hypothesis Space	NPTEL Video

Problem based learning on Bayes Theorem



Problem Based Learning on Bayesian belief Network

NPTEL Video on Sample Complexity for Finite and Infinite Hypothesis Space	
	

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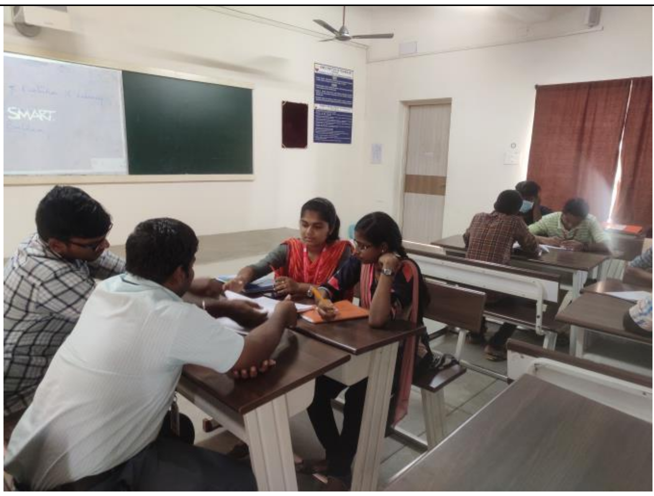
Academic Year 2021 – 2022 (Odd Semester)

Active Learning Description for Unit IV

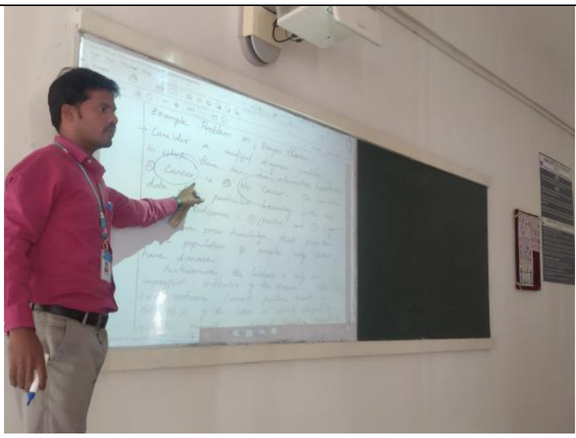
Course Code & Title: CS8082 Machine Learning Techniques
Degree, Semester & Branch: VII Semester B.E. ECE B
Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic	Name of the Innovative Practice
Remarks on KNN Algorithm	Discussion
Problems discussed in KNN	Problem based learning

Discussion on remarks on KNN Algorithm



Problem Based Learning on KNN



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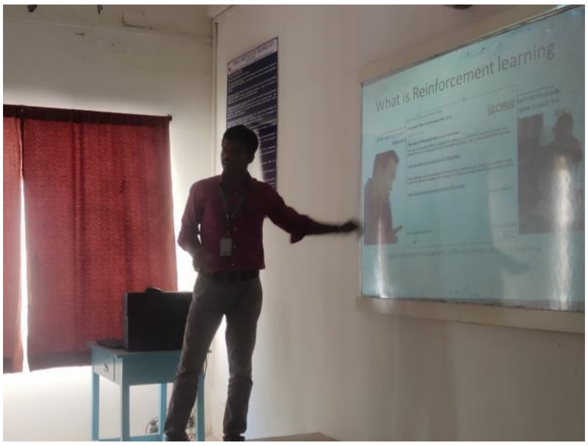
Academic Year 2021 – 2022 (Odd Semester)

Active Learning Description for Unit V

Course Code & Title: CS8082 Machine Learning Techniques
Degree, Semester & Branch: VII Semester B.E. ECE B
Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic	**Name of the Innovative Practice**
Reinforcement Learning	Inquiry based learning
Q-Learning	Problem based learning

Inquiry based learning on Reinforcement Learning	
	
Problem Based Learning on Q Learning